

usb_joystick_routed

Design Report

Rev 1 | 2026-06-17 | jlcpcb

kicad-tools 0.14.0

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	19 (17 SMD, 2 THT, 0 other)
Nets	16
Traces	570 segments
Vias	17
Board Size	80.0 x 60.0 mm

Design Overview

Theory of Operation

USB Joystick Controller

USB game controller with analog joystick

Demonstrates autolayout functionality

Communication Interfaces

Protocol	Signals
USB	USB_D+, USB_D-, VBUS

Power Architecture

Power Rails: GND, PWR_FLAG, VCC

Assembly Notes

1 fine-pitch component

- **Fine-pitch components:** 1 (U1)

Hand-Solder (THT) Components

The following 1 through-hole component is **excluded from the SMT pick-and-place file** and must be hand-soldered (or wave/selective-soldered) after SMT assembly. It appears in the BOM for sourcing.

Value	Package	Qty	References
Joystick	Joystick_Analog	1	J2

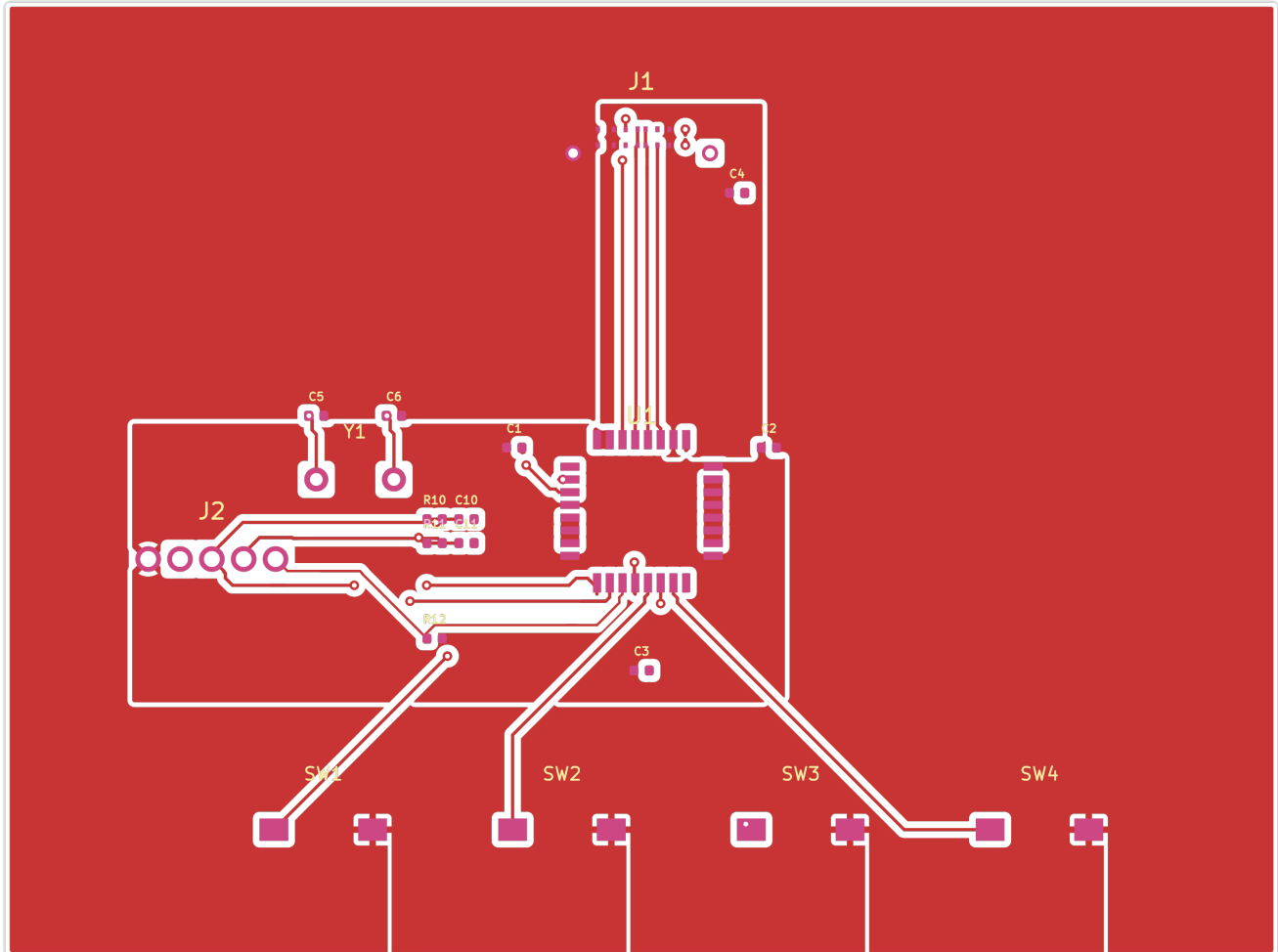
ERC Status

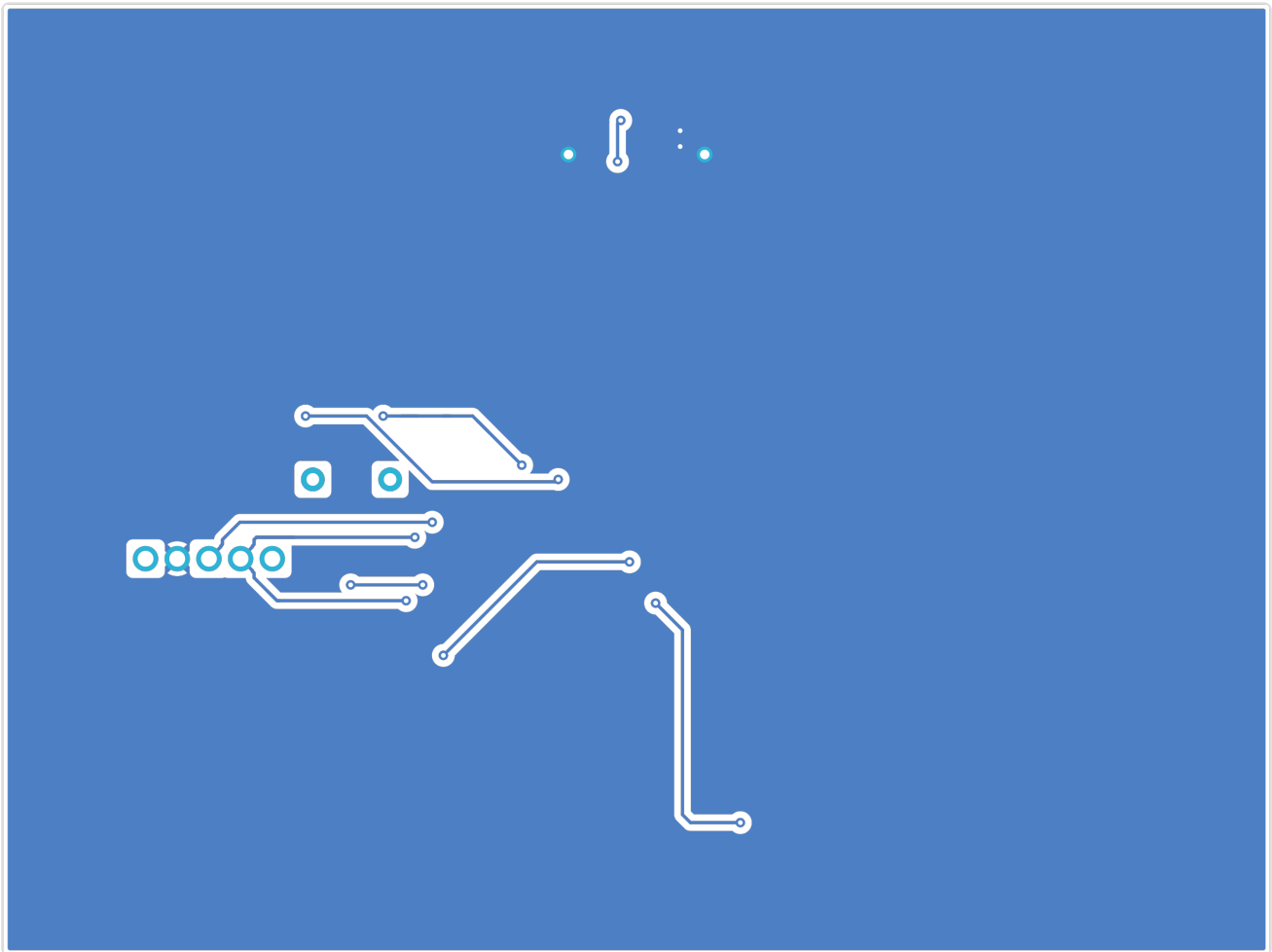
Metric	Count
Errors	0
Warnings	0

Status: SKIPPED -- ERC skipped by user request

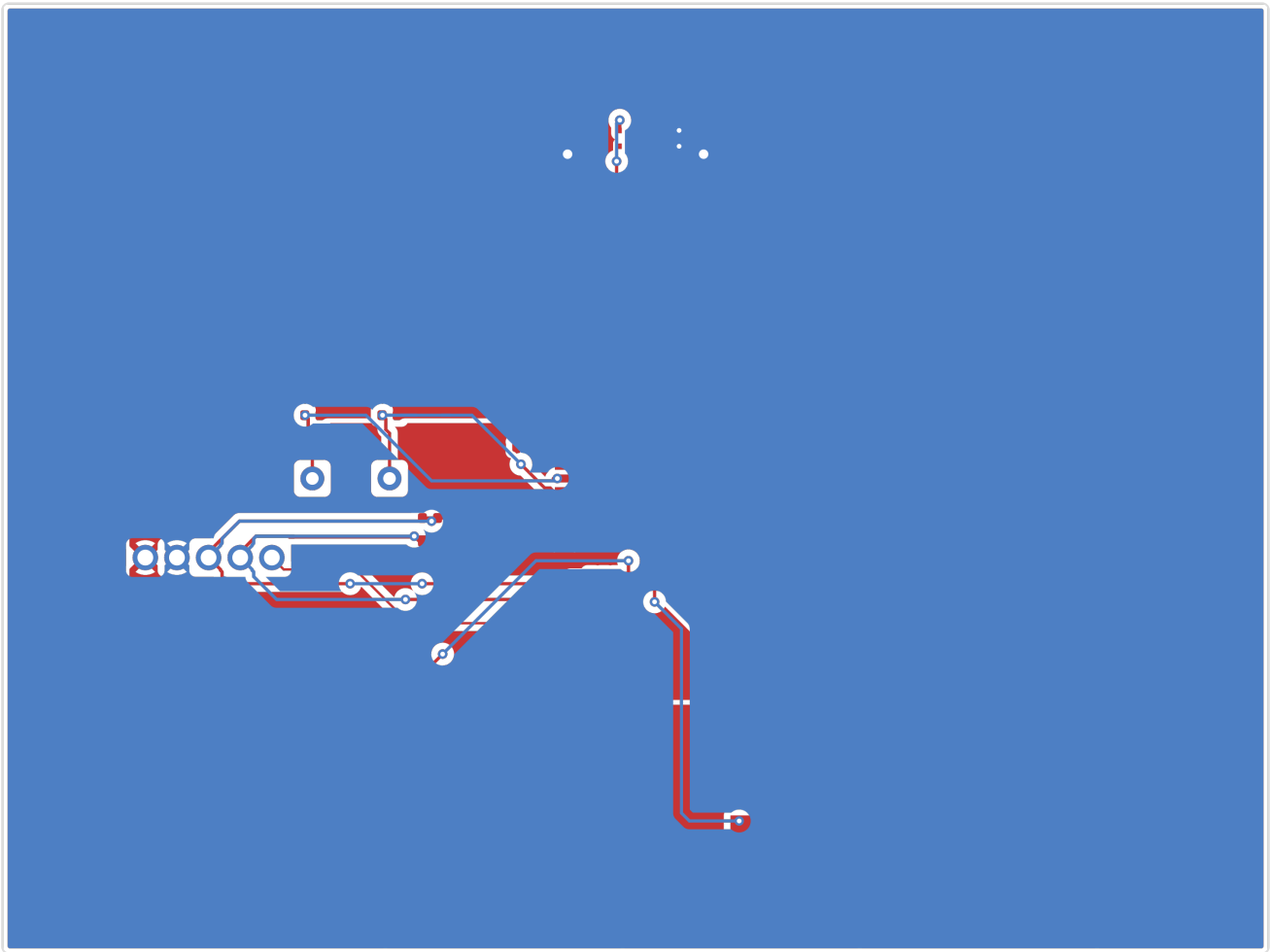
PCB layout of a USB Joystick Controller. The board is populated with a USB Type-A connector, a USB-to-serial bridge (CC2521), a microcontroller (ATmega328P), a 1.8MHz crystal, and various passive components like capacitors and resistors. The layout includes a USB shield, a joystick module, and a push button. Power and ground planes are clearly defined, with labels for VCC, GND, and VBUS. The board is labeled "USB Joystick Controller" and "KiCad E.D.A. 10.0.3".

PCB Layout

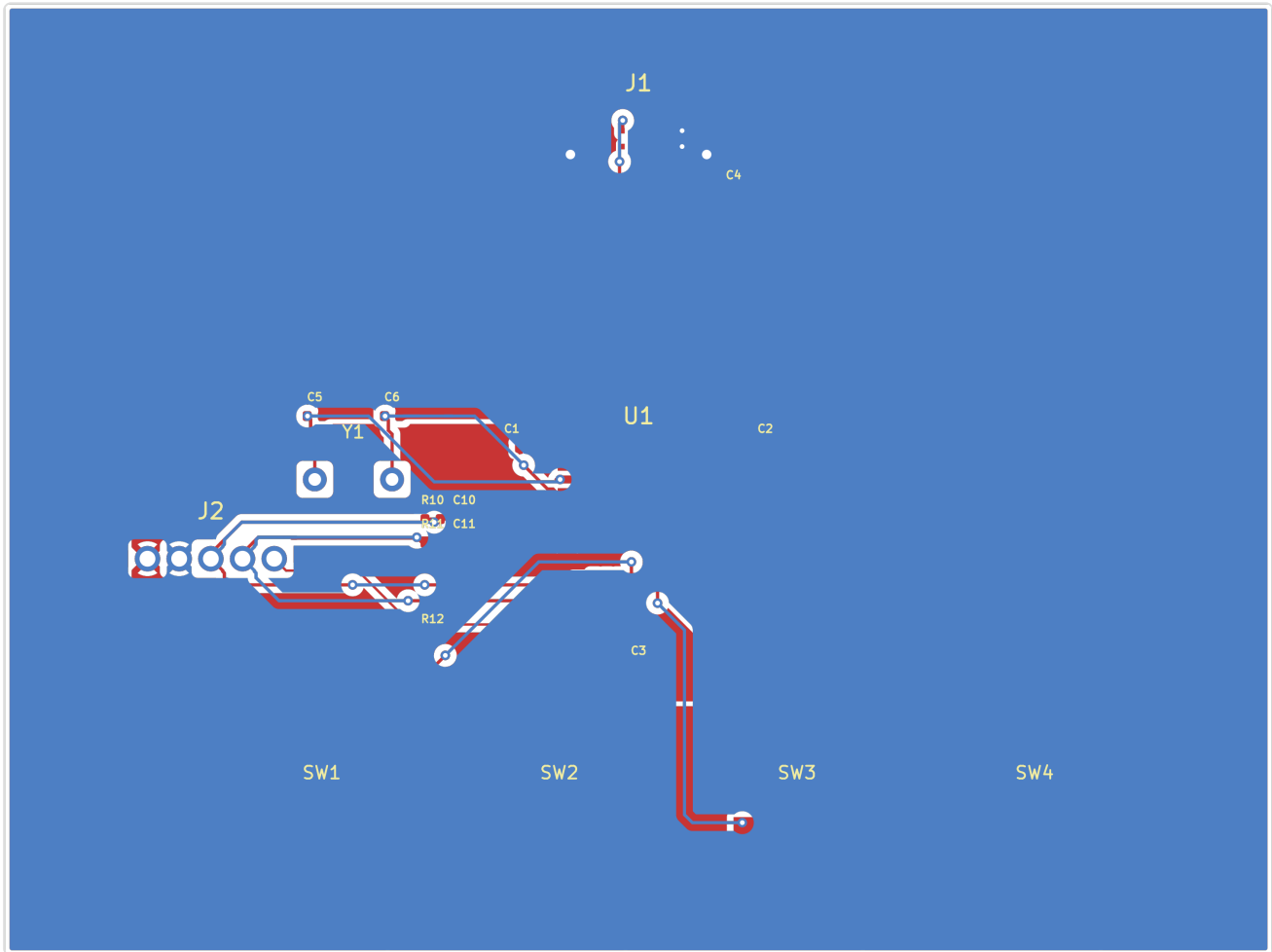




Copper

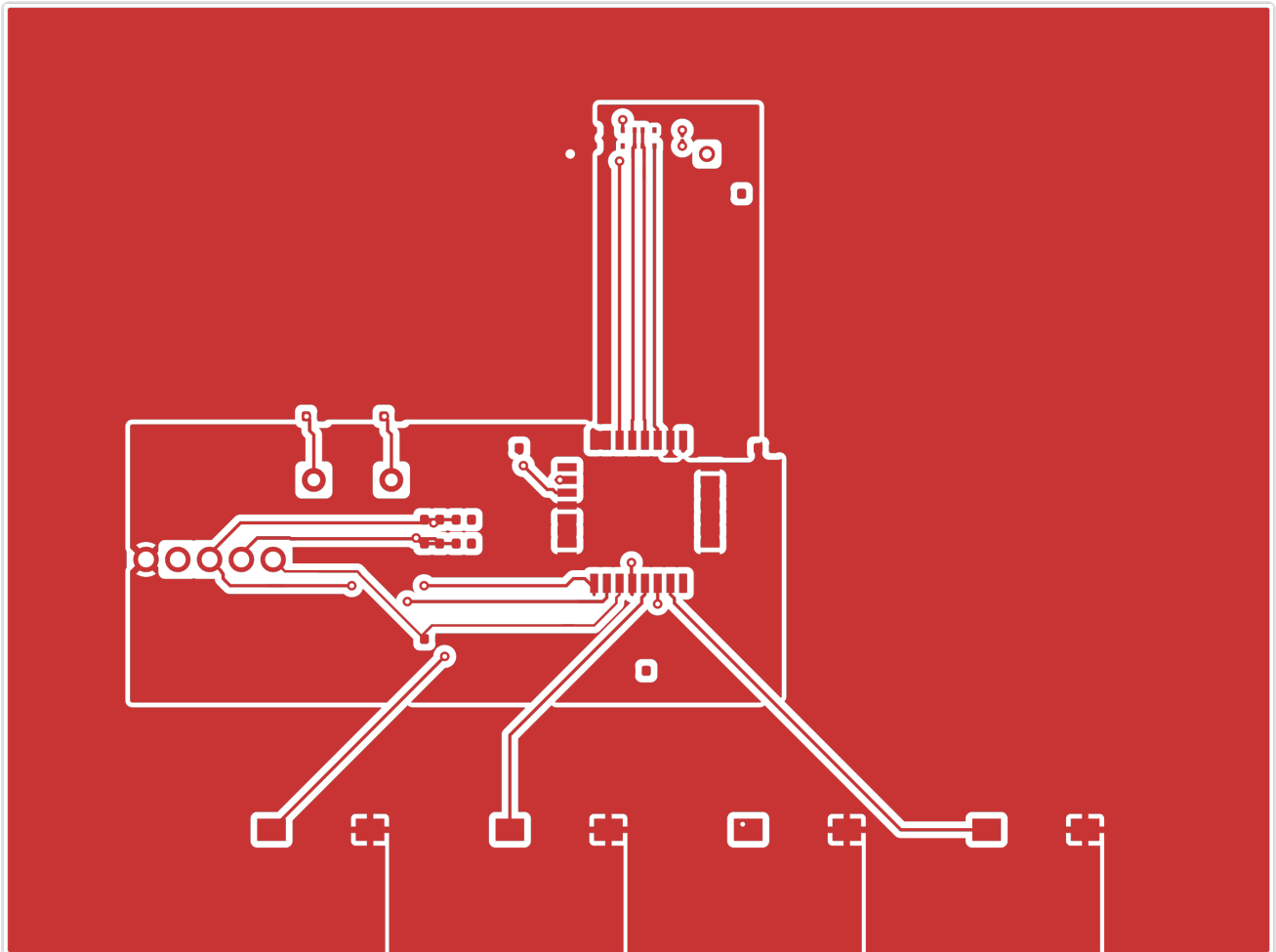


Assembly

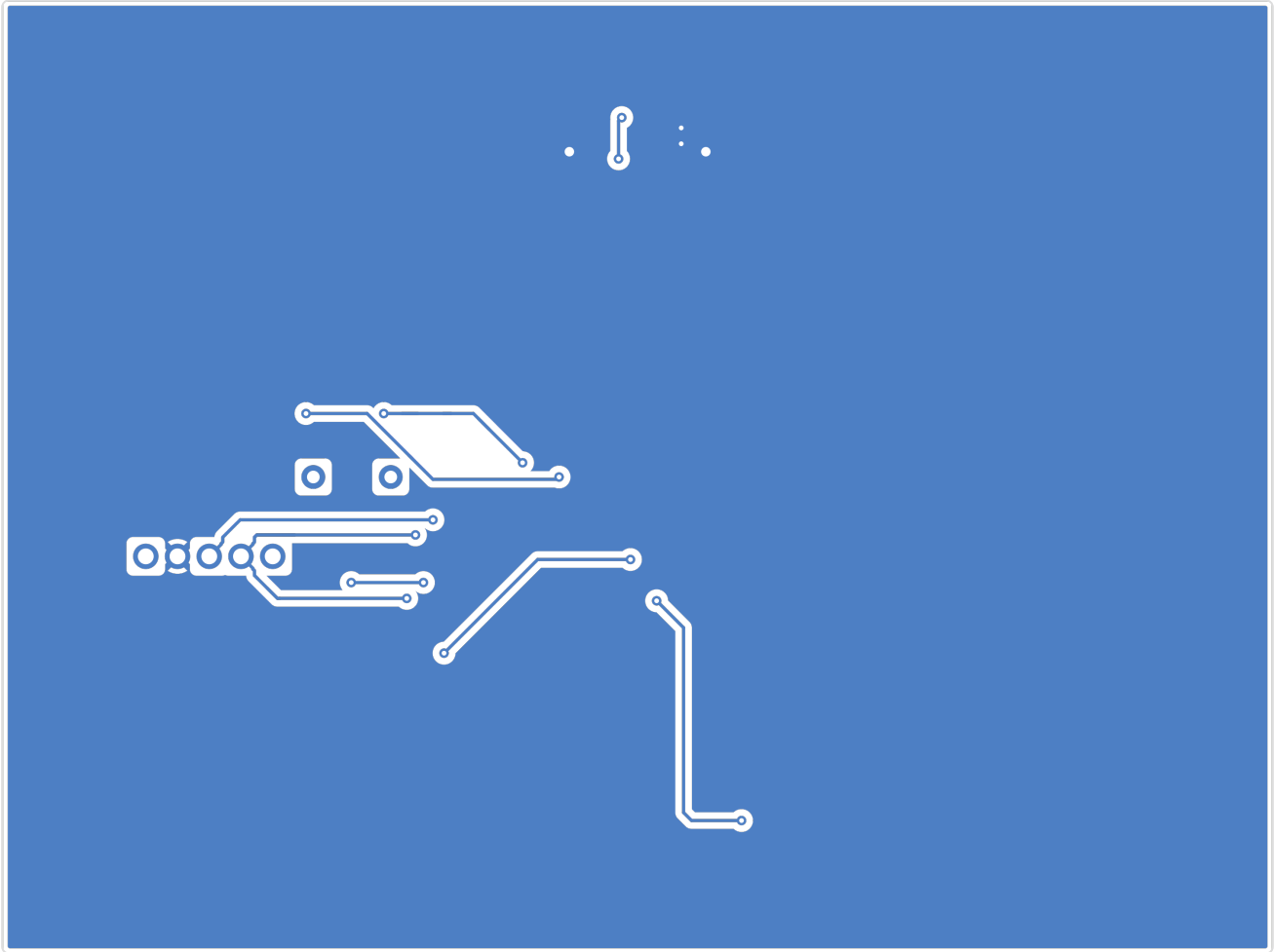


Copper Layers

F.Cu



B.Cu



Bill of Materials

Value	Package	Qty	References
100nF	C_0402_1005Metric	4	C1, C2, C3, C4
16nF	C_0402_1005Metric	2	C10, C11
22pF	C_0402_1005Metric	2	C5, C6
Joystick	Joystick_Analog	1	J2
TYPE-C	USB_C_Receptacle_GC T_USB4105	1	J1
10k	R_0402_1005Metric	3	R10, R11, R12
Button	SW_SPST_TL3342	4	SW1, SW2, SW3, SW4
MCU	TQFP-32_7x7mm_P0.8 mm	1	U1
16MHz	Crystal_HC49-U_Vertical	1	Y1

DRC Status

Metric	Count
Errors	5
Warnings	16
Blocking	5

Status: FAIL

Violations by Type

Violation Type	Count
silkscreen_text_height	16
via_in_pad	4
kicad-cli:clearance	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 5 blocking DRC violations (via_in_pad (4); kicad-cli: clearance (1))
- **[OPTIONAL]** Verify zone fill in KiCad for 3 zone-connected nets
- **[OPTIONAL]** Review 16 DRC warnings

Routing Status

Metric	Value
Signal Net Completion	100.0% (13/13)
Overall Completion	100.0%
Complete Nets	16 / 16
Zone-Connected Nets	3
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- GND
- VBUS
- VCC

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~1.36 USD
Components (estimated)	~1.33 USD
Assembly (estimated)	~2.03 USD
Total (estimated)	~4.72 USD
Batch Quantity	5
Batch Total (estimated)	~23.61 USD